THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS TWO-CHAPTER TEST SET C

Chapters 11 & 12: **Areas Related to Circles, Surface Areas and Volumes** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Use $\pi = \frac{22}{7}$ unless stated otherwise.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The area of a sector of radius 6 cm and angle 120° is (Use $\pi = 3.14$): 1

(a) 37.68 cm² **(b)** 18.84 cm² **(c)** 12.56 cm² **(d)** 75.36 cm²

2. The arc length of a sector of radius 14 cm and angle 45° is: 1

(a) 44 cm **(b)** 22 cm **(c)** 5.5 cm **(d)** 11 cm

3. An arc of length 22 cm in a circle of radius 21 cm subtends an angle of: 1

(a) 90° **(b)** 120° **(c)** 60° **(d)** 30°

4. A student states: "Doubling the angle of a sector (same radius) doubles its area." The correct response is: 1

(a) Correct **(b)** Incorrect
(c) Area becomes 4 times **(d)** Cannot be determined

5. For the area of a semicircle of radius 14 cm, Student A says 308 cm²; Student B says 616 cm². Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

6. The volume of a sphere of radius 3 cm is: 1

(a) 36π cm³ **(b)** 27π cm³ **(c)** 12π cm³ **(d)** 18π cm³

7. The curved surface area of a cone of radius 6 cm and height 8 cm is: 1

(a) 36π cm² **(b)** 60π cm² **(c)** 96π cm² **(d)** 48π cm²

8. The total surface area of a solid hemisphere of radius 7 cm is: 1

(a) 616 cm² **(b)** 462 cm² **(c)** 308 cm² **(d)** 154 cm²

9. A student states: "Joining two hemispheres of radius r gives a solid of surface area $6\pi r^2$." The correct response is: 1

(a) Incorrect — a sphere has area $4\pi r^2$ **(b)** It is $3\pi r^2$
(c) Correct **(d)** Cannot be determined

10. A toy is a cylinder (radius 1, height 7) with a hemisphere (radius 1) on top. Student A says its volume is $7\pi + \frac{2}{3}\pi$; Student B says $7\pi + \frac{4}{3}\pi$. Then: 1

(a) Student B is correct **(b)** Student A is correct
(c) Both are correct **(d)** Both are wrong

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. Find the area of a sector of radius 7 cm with angle 72°. 2

12. A pendulum 35 cm long swings through 36°. Find the length of the arc described. 2

13. A solid is a hemisphere of radius 7 cm with a cylinder of the same radius and height 5 cm on it. Find its volume. 2

14. A student says that the surface area of a cuboid formed by placing three cubes of edge 2 cm in a row is 72 cm². Justify or refute. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Find the area of the minor segment of a circle of radius 10.5 cm when the sector angle is 90°. 3

16. From a solid cylinder of radius 6 cm and height 14 cm, a hemisphere of radius 6 cm is scooped out from the top. Find the volume of the remaining solid. 3

17. A pie-chart circle of radius 7 cm has a sector showing 30% of the total. Find its angle and area. 3

OR

Find the area of the minor segment of a circle of radius 5 cm when the chord subtends 90°. (Use $\pi = 3.14$) 3

18. A toy is a cylinder of radius 3 cm and length 10 cm with a cone of height 4 cm at each end. Find its volume. 3

OR

A solid is a cone of radius 4.2 cm and height 5.6 cm on a hemisphere of the same radius. Find its total surface area. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A garden sprinkler waters a sector of radius 14 m and angle 150°. 5

(i) Find the area watered.
(ii) Find the length of the curved boundary.
(iii) Find the extra area watered if it covers the full circle.

OR

A ring-shaped flower bed lies between two concentric circles of radii 7 m and 14 m. A part of it is a sector of angle 40°. Find 5

(i) the area of this part
(ii) its perimeter.

20. **Case study:** A test tube is a cylinder of radius 1 cm and height 10 cm with a hemispherical bottom. 5

(i) Find its capacity.
(ii) Find its outer surface area.
(iii) How many such tubes can be completely filled from 1 litre?

OR

A boiler is a cylinder 2 m long with hemispherical ends, its diameter being 1 m. Find 5

(i) its volume
(ii) its surface area.

END OF QUESTION PAPER — Total: 40 Marks